



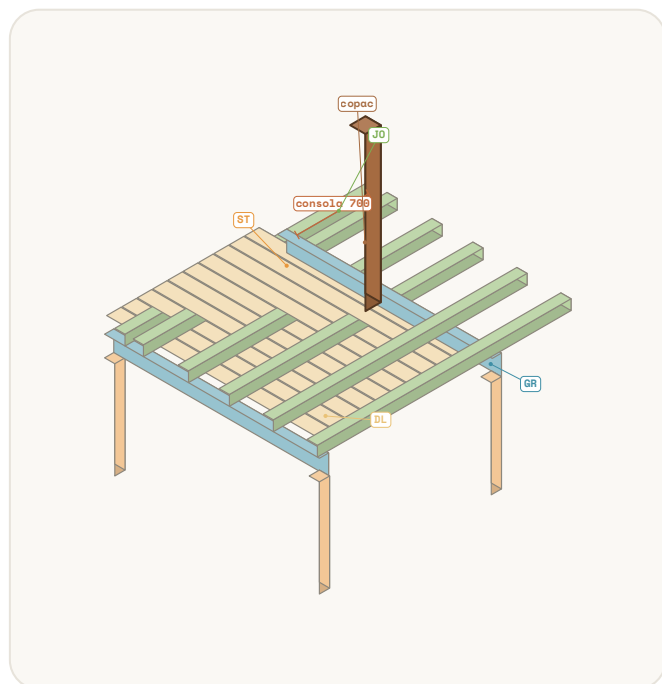
MANUAL DE CONSTRUCTIE · FAZA 1

Casuta din copac

Platforma pe patru stalpi, langa corcodusul din gradina —
pas cu pas, pentru Tudor si tata.

Ce construim

O casuta ridicata la 2,2 m pe patru stalpi (nu in copac — corcodusul e prea subtire ca s-o tina). In spate, langa gard, partea inchisa; in fata, un balcon deschis care iese 700 mm peste stalpi. Copacul trece prin podeaua balconului si il retezam ca masuta. **Numerotam stalpii: S1-S2 = spate (raman lungi, 4 m), S3-S4 = fata (se taie la +1872).**



Inaltime podea	2200 mm
Cadru stalpi	2100 × 1780 mm
Balcon (consola)	+700 mm
Grinzi	glulam 90×200
Joiste	6 × 100×100
Balustrada	1 m, goluri sub 9 cm
Copac	corcodus Ø160, masa

CODUL CULORILOR (ACELASI IN TOT MANUALUL)

ST stalp
 GR grinda
 JO joista
 DL dusumea
 PO polita
 C1/C2 coltar
 BL blocaj
CF contrafisa
 — copac

REGULA 1

Copacul nu tine nimic.
Corcodusul trece liber prin podea, cu joc. Tot ce e portant sta pe cei 4 stalpi.

REGULA 2

Balustrada, goluri sub 9 cm,
mana curenta la 1 m —
podeaua e la 2,2 m.

REGULA 3

Nimeni pe consola (balcon)
inainte de a monta
contrafisele de sub nas.

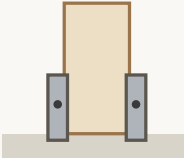
Cuprins

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Materialie

Toate piesele, cu codul lor (acelasi cod in toate desenele). Ce ai deja, ce e in comanda Hornbach si ce iei din offcut.

LE AI DEJA

			
ST 4	J0 4	PAP 4	F2 1
Stalp KVH 100x100 · stalpi KVH 100x100	Joista 100x100 · joiste (din lotul de 6)	Papuc reazem U · papuci, deja in beton	Grund Kober 4 L · grund Kober 4 L

IN COMANDA HORNBACH (4.101 LEI)

			
GR 2	J0 2	DL 17	C1 4
Grinda glulam 90x200 · grinzi glulam 90x200	Joista 100x100 · completare joiste	Dusumea larice 28x145 · dusumea larice	Coltar 100x100x90 · coltare 100x90
			
C2 14	H1 1 pac	H2 1 pac	H3 1 pac
Coltar 90x90x65 · coltare 90x65: 12 joiste + 2 grinda spate	Heco 8x200 · Heco 8x200	Heco 6x100 · Heco 6x100	Heco 6x80 · Heco 6x80
			
H4 3 pac	B1 1	B2 1 pac	B3 1 pac
Inox 5x60 · inox 5x60 (204 necesare)	Tija M12 (taie 220) · tija M12 (taie 4x ~220)	Bulon M12x120 · M12x120 baza	Saiba M12 · saibe M12



P **B4** 1 pac
Piulita M12 · piulite
M12



P **F1** 2
Ulei tec · ulei tec

DIN OFFCUT



P **P0** 2
Polita (offcut) ·
polite



P **BL** ~12
Blocaj (offcut) ·
blocaje



P **RM** 2
Traverse rama ·
traverse rama copac



P **CF** 6
Contrafisa (offcut) ·
contrafise



P **PT** ~6
Proptea temporara ·
proptele

BALUSTRADA (FAZA 1)



P **SB** ~7
Stalpisor 58x58 ·
stalpisor balustrada
58x58



P **SP** ~24
Sipci balustrada ·
sipci 18x18, gol <9 cm



P **MC** ~7 m
Mana curenta · mana
curenta



M10

P **BM** ~12
Bulon M10 · bulon M10
balustrada

DE ALES (NU OPRESTE STARTUL)



P **F3** 1
Topcoat colorat ·
topcoat colorat exterior

Scule de care ai nevoie

Pregateste-le pe toate inainte de start. Pe fiecare fisa apar doar sculele etapei.

Boloboc / nivela cu furtun

verticalitate + orizontalitate

Bormasina + biti

suruburi Heco si gauri

Burghiu lemn 13 mm

gauri pentru buloane M12

Cheie 19 (M12)

buloane baza + polita

Cheie M10

balustrada

Bomfaier

taiat tija filetata

Fierastrau / circular

taieri stalpi, dusumea, offcut

Echer mare

unghiuri drepte

Ruleta + creion

masuratori, trasaj

Cui 5 mm

distantier intre scanduri

Pensula

grund + ulei

2 persoane

la grinzi si ridicari

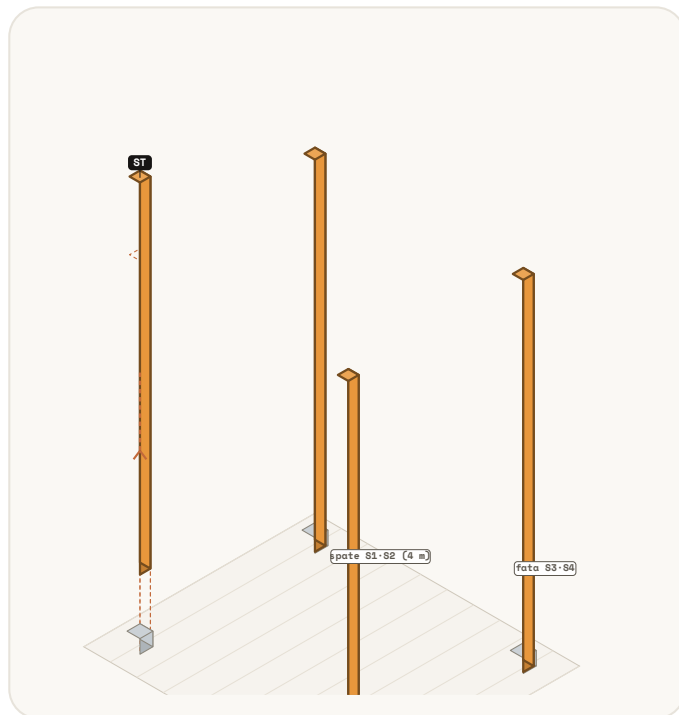
FISA 1 / 11

Stalpii in papuci

~2-3 ore

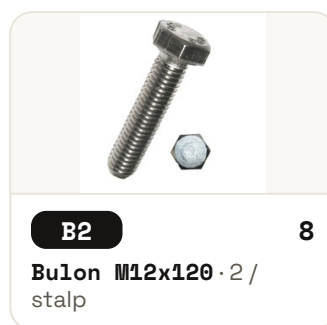
Mediu

2 persoane

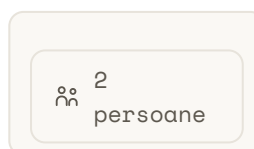
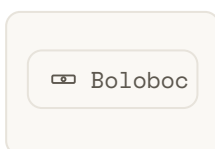


 ST stalp

PIESE



SCULE



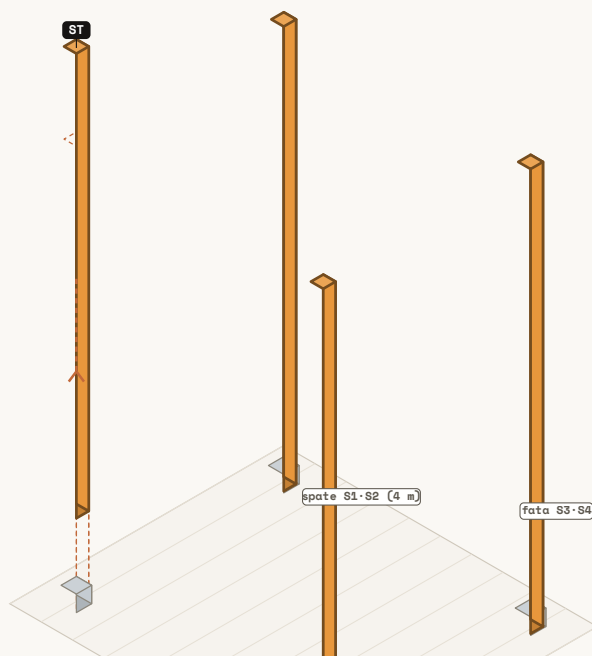
PAS CU PAS

● cap surub (fata) ○ ascuns (pe spate) → directie **oblic** = la unghi

Folosim **suruburi + buloane**, nu cuie.

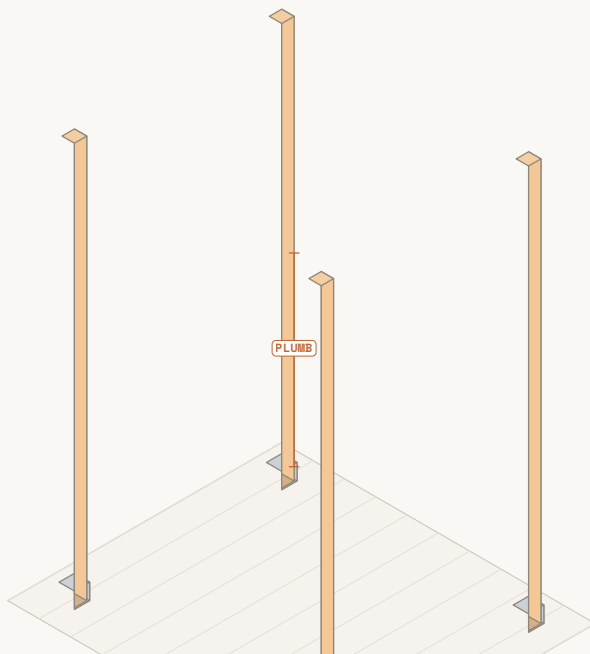
1

Aseaza toti 4 stalpii in papucii lor din beton, coborati drept in U. TOTI raman lungi acum — fata o tai la Fisa 2.



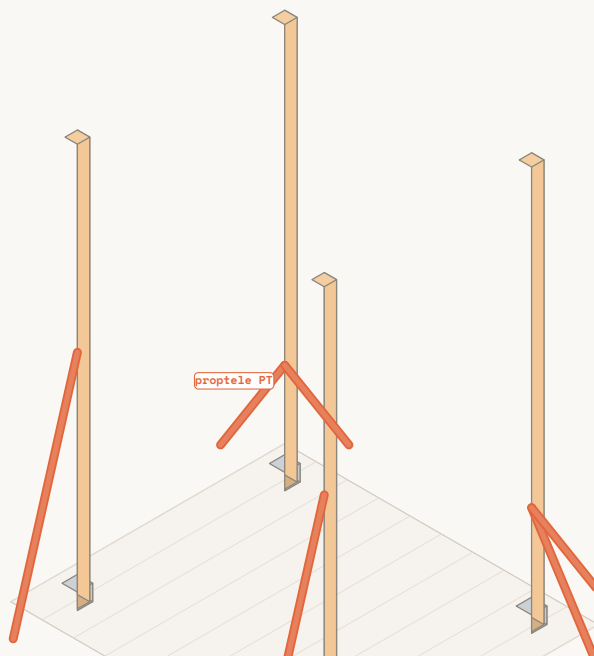
2

Adu fiecare stalp perfect vertical (la plumb), verificat pe doua fete cu bolobocul.



3

Sprijina fiecare stalp cu 2 proptele la baza. La cei 2 stalpi inalti din spate, pune si cate o proptea spre varf.

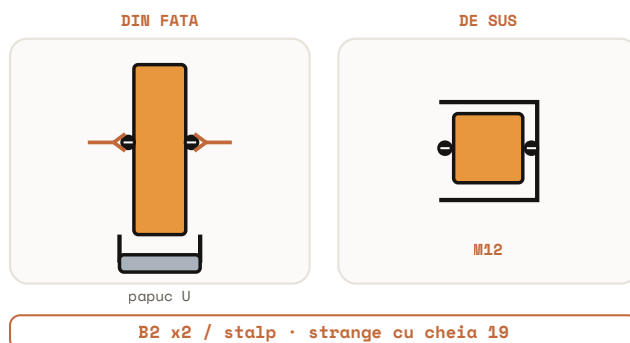


! CRITIC — Verifica fiecare stalp la plumb pe 2 fete inainte de a strange proptelele.

4

Prinde suruburile B2 in papuc DOAR slab (2 / stalp) — sa poti inca corecta. Le strangi la Fisa 2.

ZOOM · CUM SE PRINDE



GATA CAND

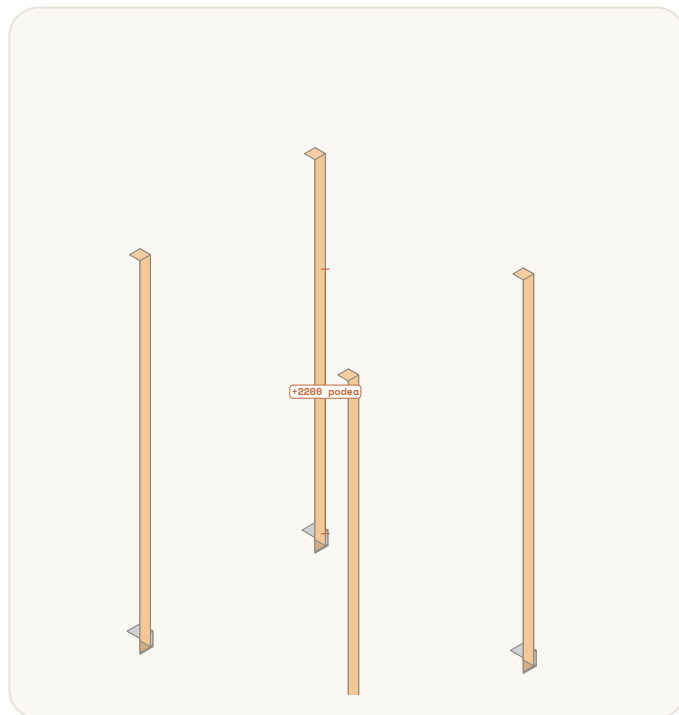
- ☒ Toti 4 stalpii verticali pe 2 fete
- ☒ Proptele la fiecare stalp
- ☒ Suruburi inca slabe

FISA 2 / 11

Nivel +2200 si taiere stalpi fata

~1-2 ore

CRITIC



 ST stalp

PIESE



ST

Stalp KVH 100x100
doar cei 2 din fata se taie

SCULE

 Nivelă cu furtun / laser

 Creion

 Fierastrau

 Echer

PAS CU PAS

 cap surub
(fata)

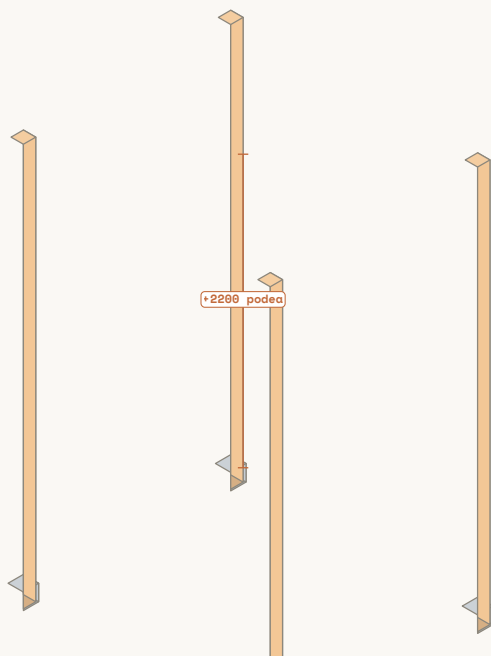
 ascuns (pe
spate)

 directie **oblic** = la
unghi

Folosim **suruburi + buloane**, nu cuie.

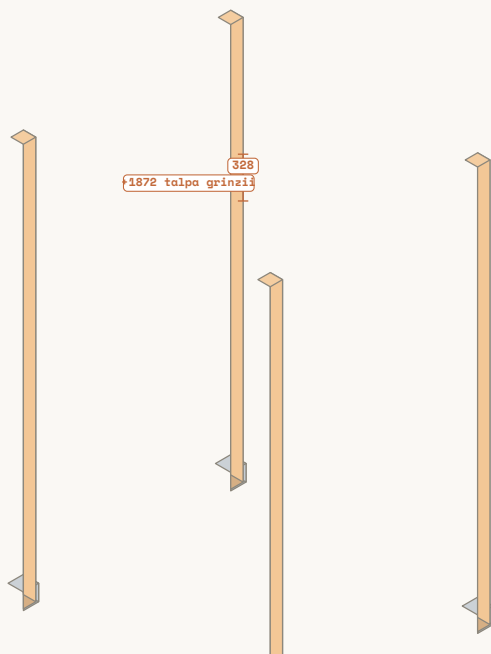
1

Marcheaza +2200 (fata podelei) pe toti 4 stalpii. Foloseste o scandura dreapta + boloboc, nu masura separat fiecare.



2

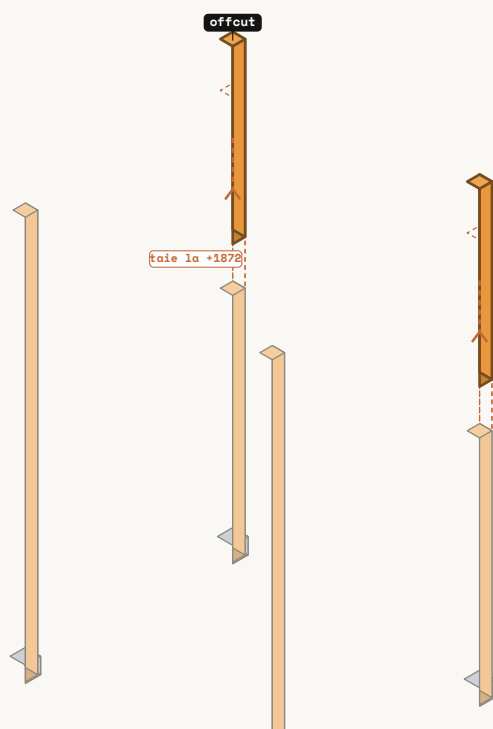
Coboara 328 mm si marcheaza +1872 — aici sta talpa GRINZII. Grinda 200 + joista 100 + dusumea 28 = 328, asa ajungi la +2200.



! CRITIC — Linia +1872 trebuie identica pe toti 4 stalpii.

3

Taie DOAR stalpii din fata (S3, S4) la +1872. Partea taiata o pastrezi pentru polite. Stalpii din spate raman INTREGI.

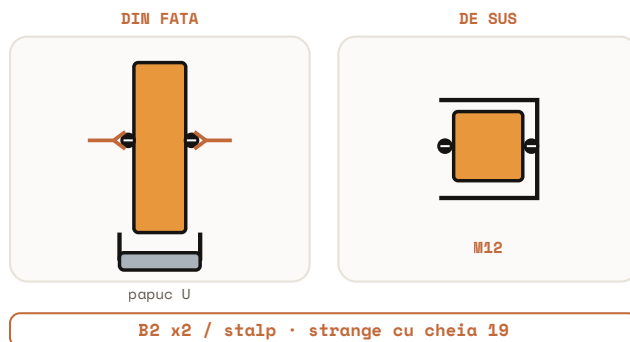


! CRITIC — Taie doar fata. Stalpii din spate raman intregi — ei sunt stalpii Fazei 2.

4

Teseste muchia taiata. Abia acum strange definitiv suruburile B2 de la baza.

ZOOM · CUM SE PRINDE



GATA CAND

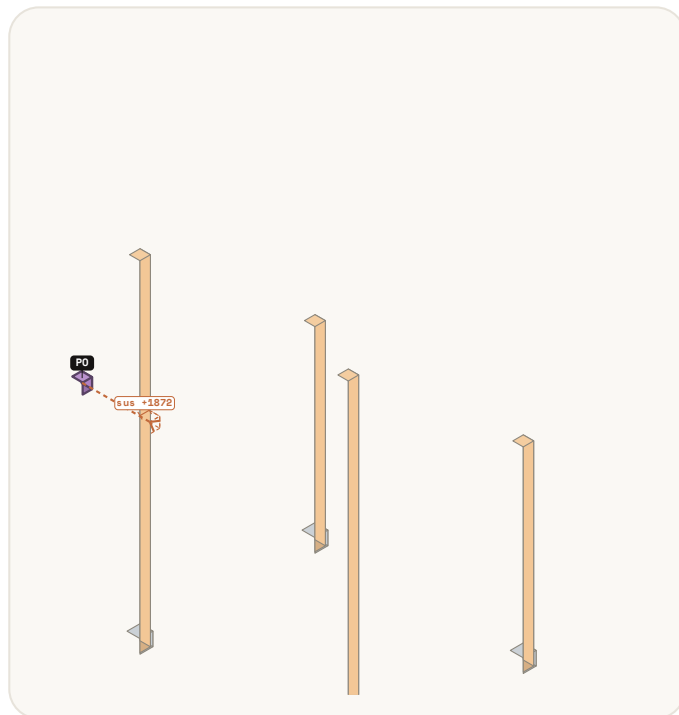
- ☐ Linia +1872 identica pe toti
- ☐ Doar stalpii fata taiati
- ☐ Stalpii spate INTREGI
- ☐ Suruburi baza stranse

FISA 3 / 11

Polite pe stalpii din spate

~1 ora

CRITIC



P0 polita

PIESE

			
P0 2	B1 4	B3 8	B4 4
Polita (offcut) · offcut 100x100	Tija M12 (taie 220) · tija ~220	Saiba M12 · 2 / bulon: cap + piulita	Piulita M12 · piulite

SCULE

 Bormasina + burghiu 13	 Cheie 19	 Bomfaier
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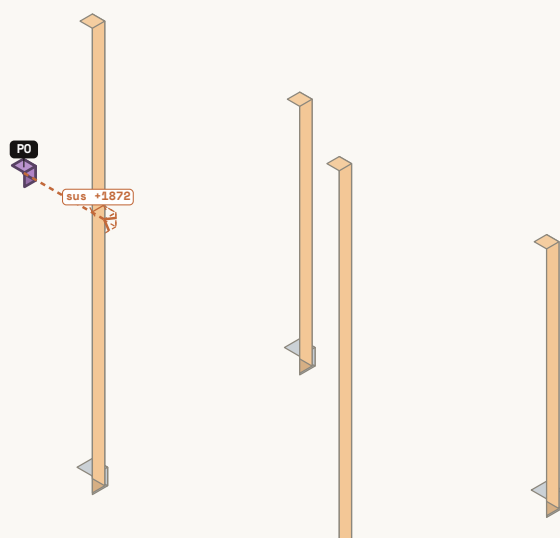
PAS CU PAS

● cap surub (fata) ○ ascuns (pe spate) → directie **oblic** = la unghi

Folosim **suruburi + buloane**, nu cuie.

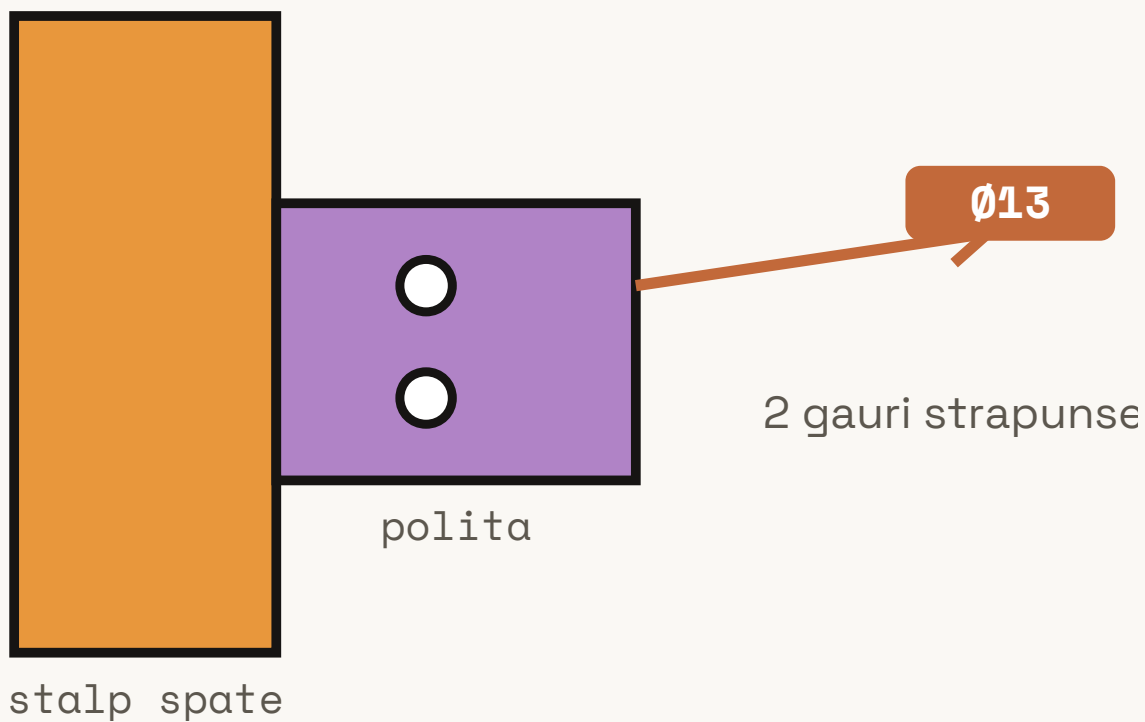
1

Aseaza polita (bloc 100x100 din offcut) pe fata interioara a stalpului din spate, cu fata de sus la +1872.



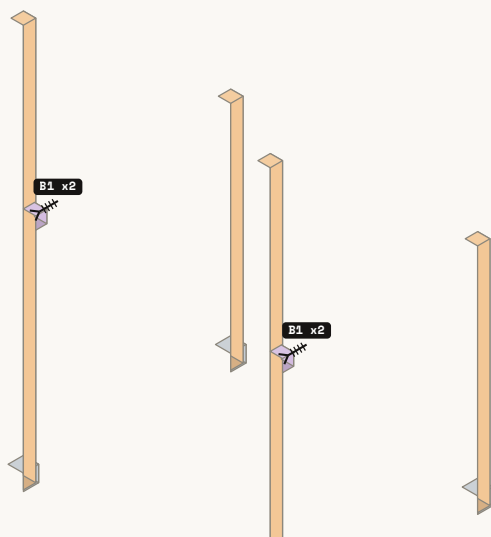
2

Gaureste prin polita in stalp: 2 gauri strapunse de 13 mm pe fiecare polita.



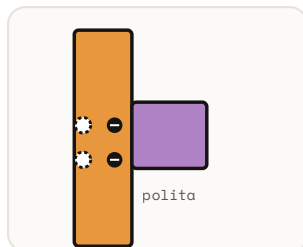
3

Bate 2 buloane M12 (B1) cu saiba (B3) prin polita in stalp; strange cu piulita (B4) pe spate. Repeta la al doilea stalp.

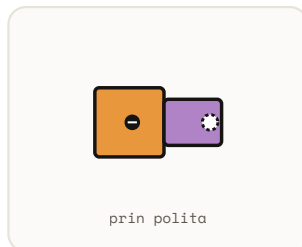


ZOOM · CUM SE PRINDE

LATERAL



DE SUS



B1 x2 / polita · cap pe fata, piulita pe spate

! CRITIC — Strange buloanele M12 pana polita nu se mai misca deloc.

GATA CAND

-  Ambele polite sus la +1872
-  2 buloane stranse / polita
-  Polita nu se misca

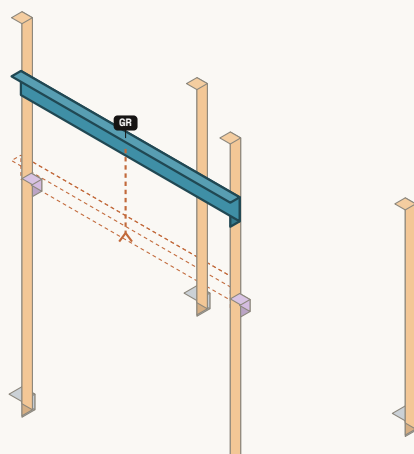
FISA 4 / 11

Grinda din spate pe polite

~1 ora

CRITIC

2 persoane

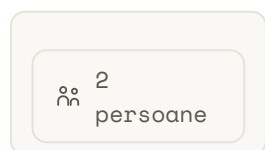


GR grinda C1/C2 coltar

PIESE



SCULE



PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



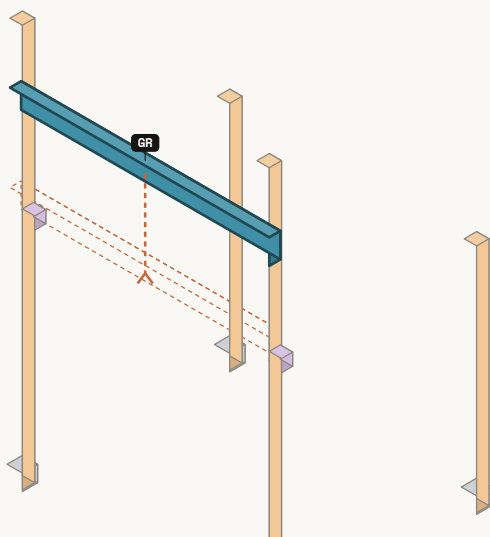
directie

oblic = la
unghi

Folosim **suruburi + buloane**, nu cuie.

1

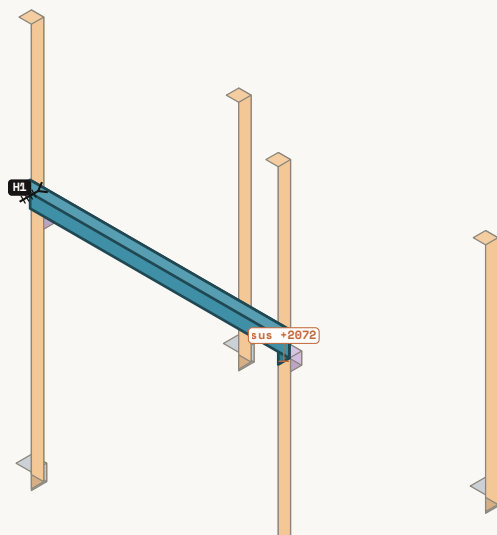
Ridicati in DOI grinda din spate si asezati-o pe cele doua polite. Sta pe polita — nu o tineti voi.



! CRITIC — Grinda sta pe polite; nu o tineti voi cat o prindeti.

2

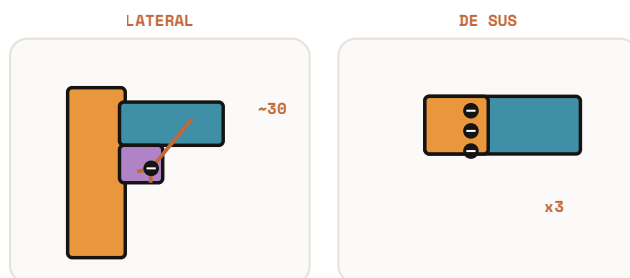
Verifica fata de sus la +2072 si orizontalitatea pe toata lungimea.



3

Prinde grinda de fiecare stalp cu 3 suruburi H1 oblice PLUS un coltar C2 anti-smulgere in stalp si in grinda. Grinda nu trebuie sa se poata ridica de pe polita.

ZOOM • CUM SE PRINDE



H1 oblic in stalp • x3 • tine grinda lipita

! CRITIC — Coltar C2 anti-smulgere la fiecare capat — grinda spate nu se poate ridica de pe polita.

GATA CAND

- ☒ Grinda pe ambele polite
- ☒ Fata sus la +2072
- ☒ Coltar anti-smulgere la fiecare capat
- ☒ Buloane polita REVERIFICATE dupa asezarea grinzii

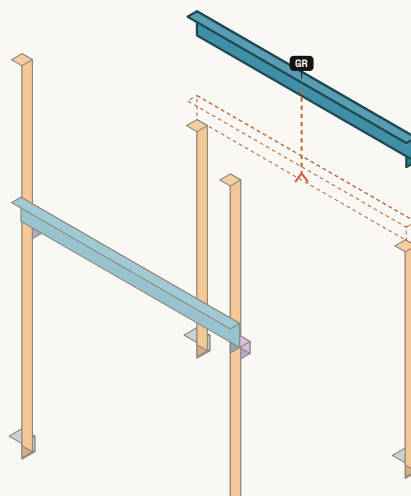
FISA 5 / 11

Grinda din fata pe varful stalpilor

~1 ora

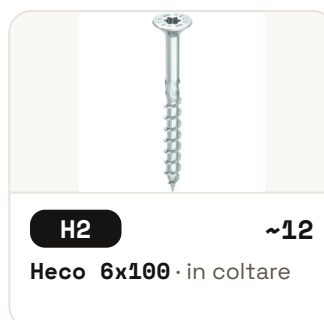
CRITIC

2 persoane

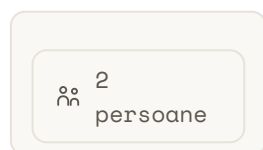


 **GR** grinda  **C1/C2** coltar

PIESE



SCULE



PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



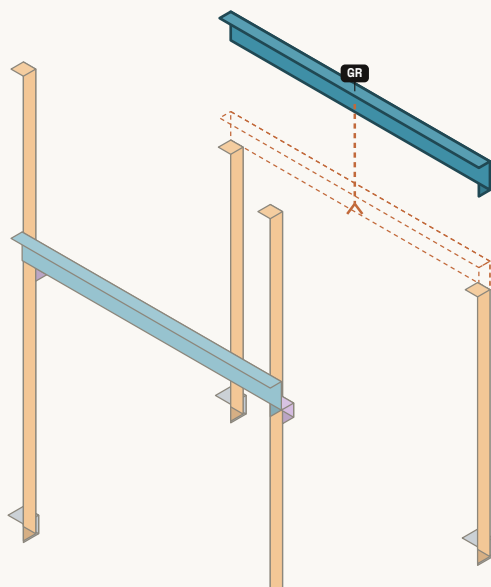
directie

oblic = la
unghi

Folosim **suruburi + buloane**, nu cuie.

1

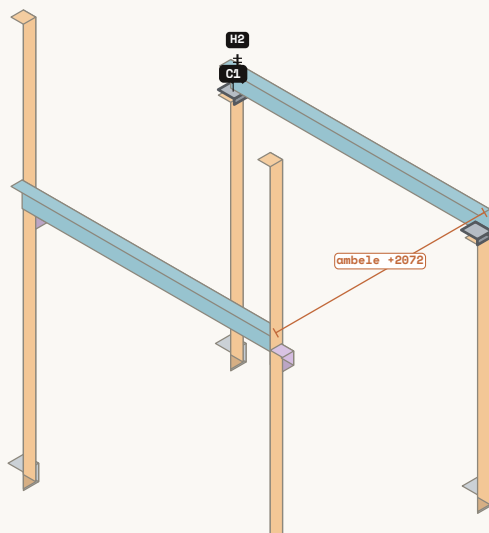
Asezati grinda din fata PE varful stalpilor S3/S4 (taiati). Sprijin direct pe lemn, la +1872.



! CRITIC — Sprijin direct pe varful taiat la +1872, perfect orizontal.

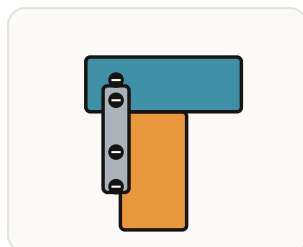
2

Prinde cu cate un coltar C1 pe fiecare fata (2/stalp) + suruburi H2 in stalp si in grinda.

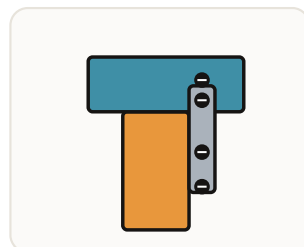


ZOOM · CUM SE PRINDE

FATA A



FATA B (opus)



C1: 2 coltare / stalp · fiecare cu 2 in stalp + 2 in grinda

3

Verifica: ambele grinzi orizontale, varful la +2072. Ai cadrul de baza.

! CRITIC — Ambele grinzi la +2072, perfect orizontale, inainte de a continua.

GATA CAND

☒ Grinda pe varful stalpilor

☒ 2 coltare C1 / stalp

☒ Ambele grinzi la +2072

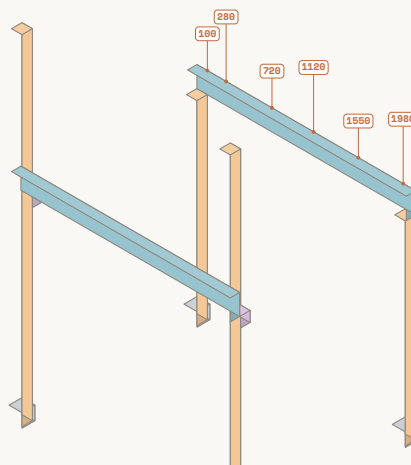
FISA 6 / 11

Cele 6 joiste

~2 ore

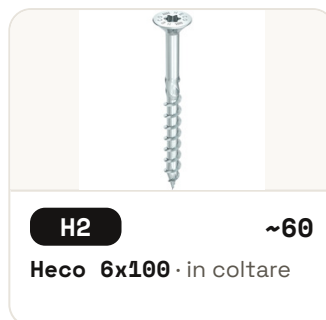
CRITIC

2 persoane



J0 joista P C1/C2 coltar

PIESE



SCULE



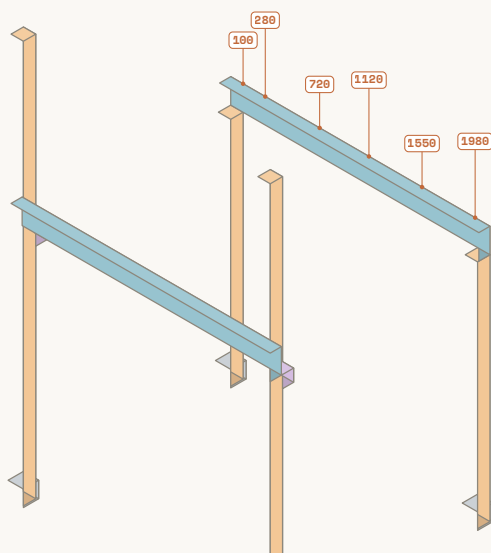
PAS CU PAS

● cap surub (fata) ○ ascuns (pe spate) → directie oblic = la unghi

Folosim **suruburi + buloane**, nu cuie.

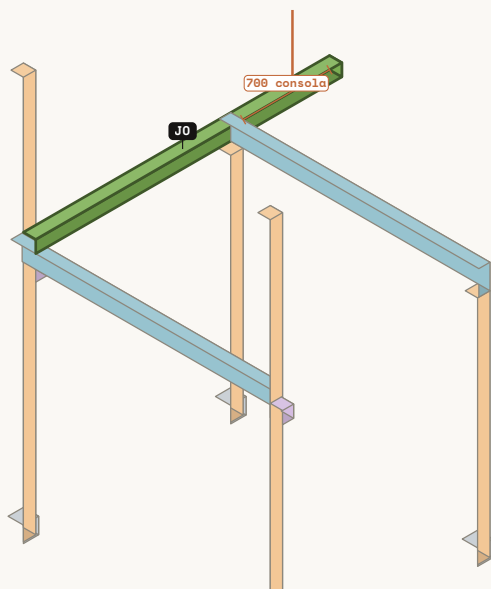
1

Marcheaza pe ambele grinzi pozitiile celor 6 joiste, de la coltul S4: 100 / 280 / 720 / 1120 / 1550 / 1980 mm.



2

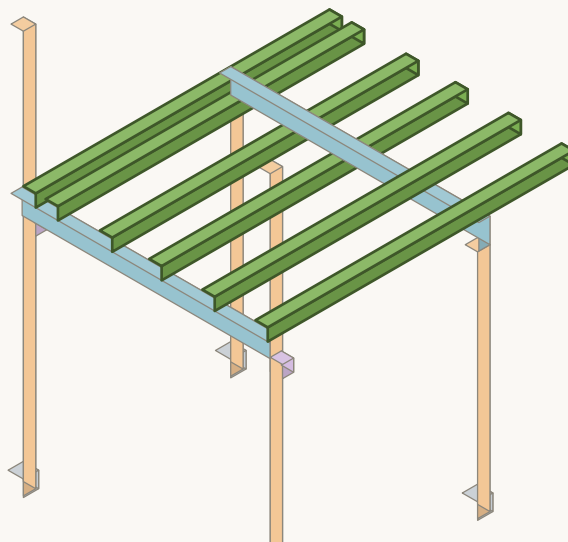
Aseaza prima joista peste ambele grinzi, la pozitia 100. Capatul din fata iese 700 mm in gol — asta e consola (balconul).



! CRITIC — Consola de 700 mm trebuie egala la toate joistele.

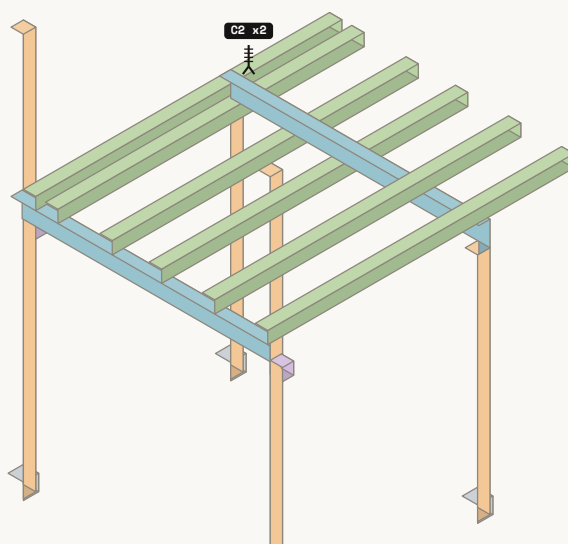
3

Aseaza toate cele 6 joiste la pozitiile marcate. Fiecare reazema pe AMBELE grinzi.



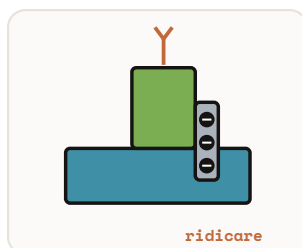
4

La FIECARE reazem (fata + spate) prinde un coltar C2 pe lateral — 12 in total. Tine joista jos la consola.

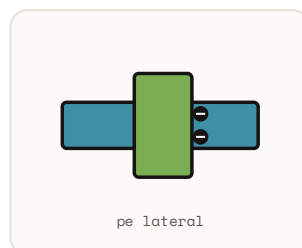


ZOOM · CUM SE PRINDE

LATERAL



DE SUS



C2 x2 / joista (fata + spate) · TOTAL 12

! CRITIC — Coltar C2 la fiecare reazem, fata si spate. Niciun reazem nelegat.

GATA CAND

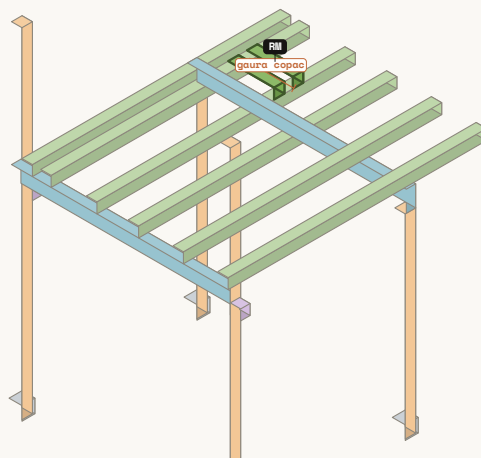
-  6 joiste la pozitii
-  Consola 700 mm
-  12 coltare C2

FISA 7 / 11

Gaura copacului si masa

~1-2 ore

Atentie

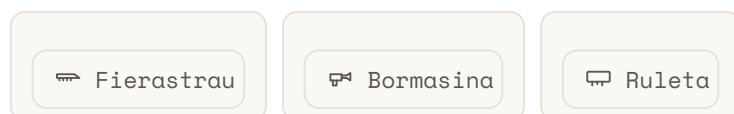


 JO joista

PIESE



SCULE

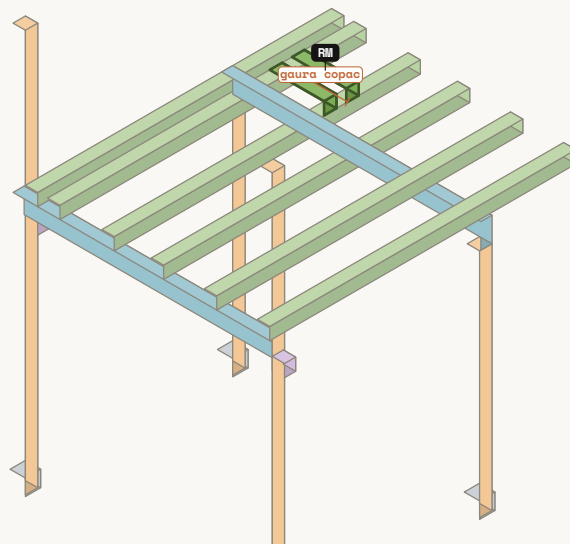


PAS CU PAS

● cap surub (fata) ○ ascuns (pe spate) → directie **oblic** = la unghi

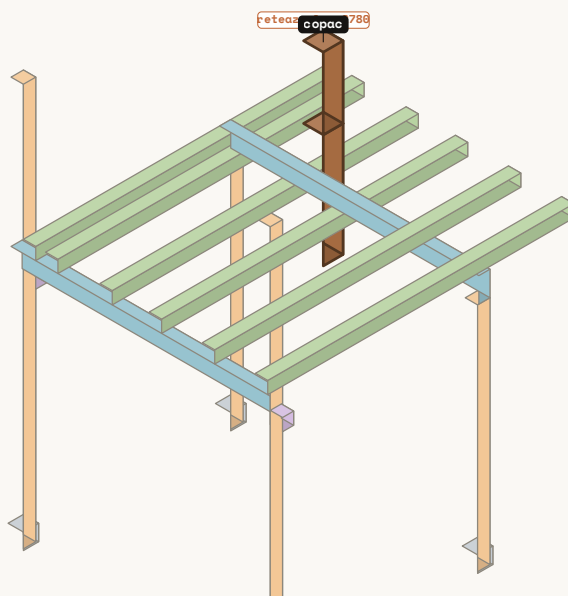
Folosim **suruburi + buloane**, nu cuie.

- 1** Intre joistele de la 280 si 720 — trunchiul cade chiar intre ele, fara sa le atinga — monteaza 2 traverse scurte (RM) care inchid rama gaurii.



2

Reteaza corcodusul la ~+2780 de la sol (~580 peste podea). Lasa joc de 3-5 cm jur-impjurul trunchiului.



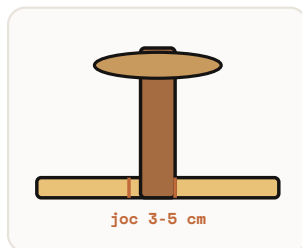
! CRITIC — Joc de 3-5 cm in jurul trunchiului — copacul se misca in vant.

3

Blatul mesei se fixeaza de podea, prin cadrul lui — NICIODATA de copac.

ZOOM · CUM SE PRINDE

LATERAL



REGULA



Blatul pe cadrul lui propriu · NICIODATA pe copac

! CRITIC — Blatul se prinde de cadrul lui propriu, NICIODATA de copac.

GATA CAND

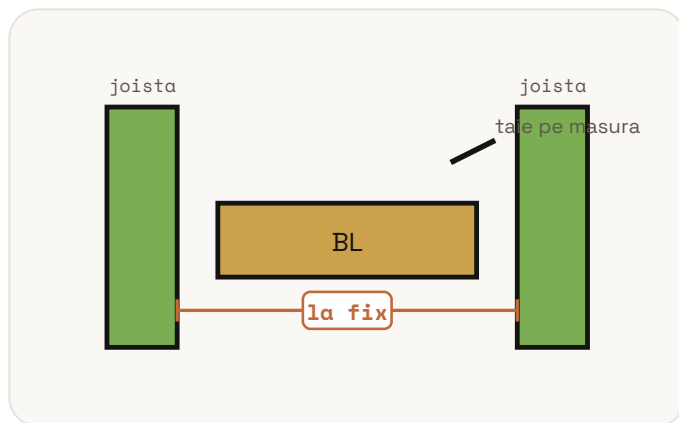
-  Rama intre joistele 280-720
-  Joc 3-5 cm in jurul trunchiului
-  Blatul NU pe copac

FISA 8 / 11

Blocaje intre joiste

~1 ora

Normal

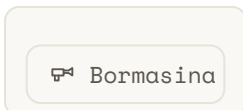


BL blocaj

PIESE



SCULE



PAS CU PAS

● cap surub
(fata)

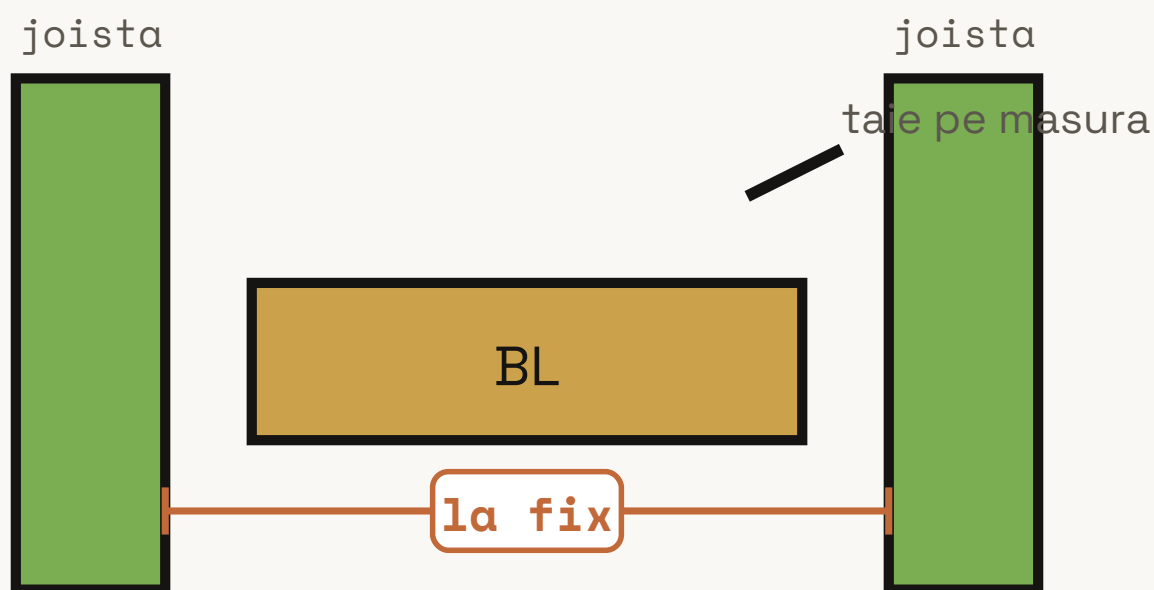
○ ascuns (pe
spate)

→ directie **oblic** = la
unghi

Folosim **suruburi + buloane**, nu cuie.

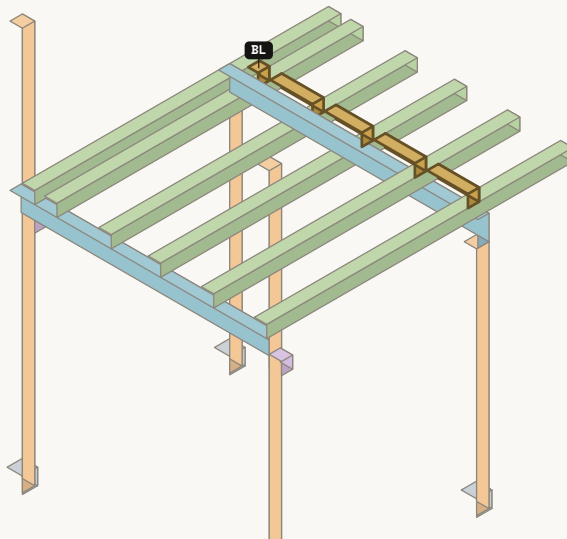
1

Taie bucati scurte de lemn (blocaje, BL) exact cat distanta dintre joiste.



2

Pune cate un blocaj intre joiste, PESTE fiecare grinda (fata si spate).

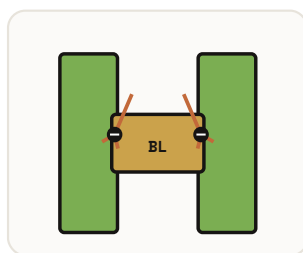


3

Prinde fiecare blocaj cu suruburi H3 oblic. Opresc joistele sa se raseasca — podeaua devine ferma.

ZOOM · CUM SE PRINDE

LATERAL



DE SUS



H3 oblic · cate 1-2 / capat

GATA CAND

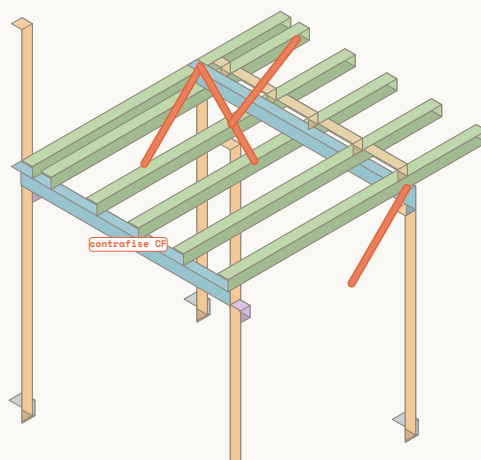
-  Blocaje peste ambele grinzi
-  Prinse oblic cu H3
-  Podeaua e ferma

FISA 9 / 11

Contravantuiri si scoatem proptelele

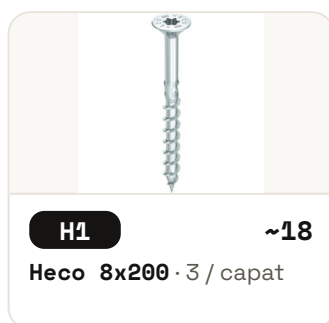
~2 ore

CRITIC

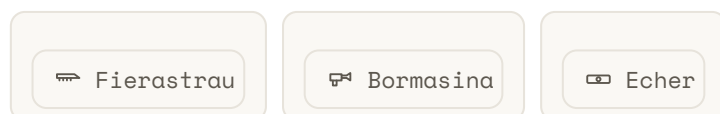


 CF contrafisa

PIESE



SCULE



PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

oblic

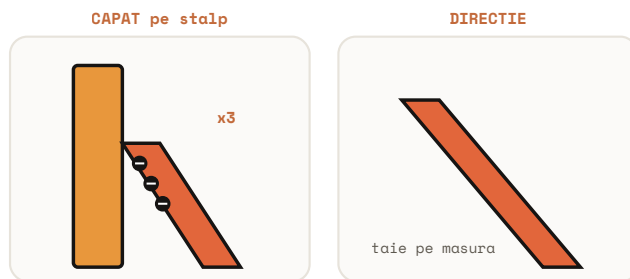
= la
unghi

Folosim **suruburi + buloane**, nu cuie.

1

Masoara si taie capatul fiecarei contrafise (diagonala) pe potrivalea.

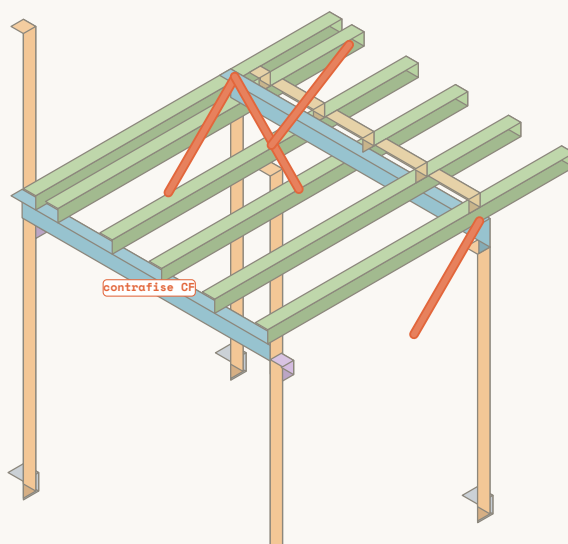
ZOOM · CUM SE PRINDE



H1 x3 / fiecare capat · fara coltar

2

Monteaza contravantuirile in planul stang, drept, fata si sub nasul consolei. La contrafisa de sub nas prinde-o pe LATERAL sau adauga un coltar — acolo lucreaza la intindere. Platforma devine rigida.



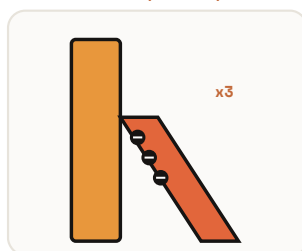
! CRITIC — Contrafisele prinse la AMBELE capete inainte de a scoate proptelele.

3

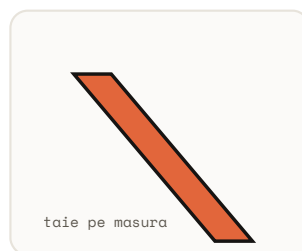
Prinde fiecare capat cu 3 suruburi H1.

ZOOM · CUM SE PRINDE

CAPAT pe stalp



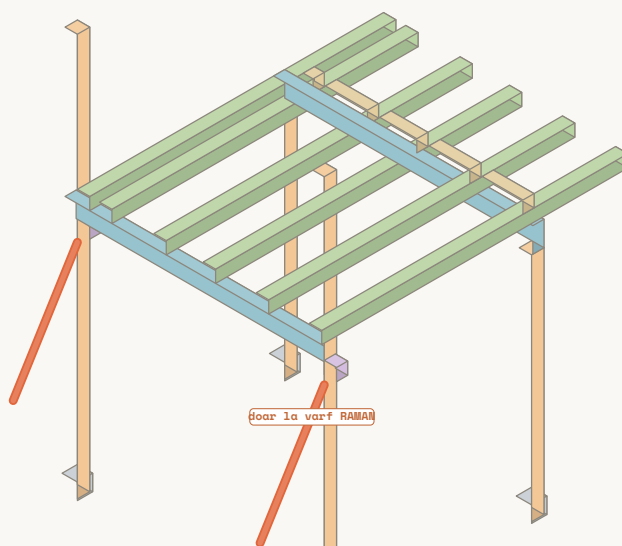
DIRECTIE



H1 x3 / fiecare capat · fara coltar

4

SCOATE proptelele de la baza. Cele de la varful stalpilor inalti din spate RAMAN pana la peretii din Faza 2.



! CRITIC — Scoti doar proptelele de la baza; cele de la varful stalpilor spate raman pana in Faza 2.

GATA CAND

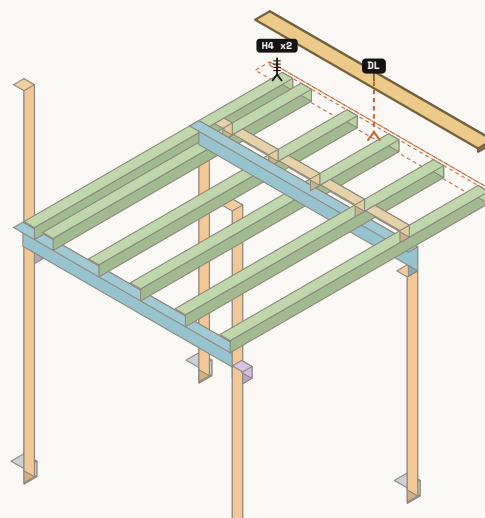
- ☐ Diagonale pe toate planurile + sub consola
- ☐ 3 suruburi / capat
- ☐ Proptele baza scoase; varf raman

FISA 10 / 11

Dusumeaua de larice

~3-4 ore

Normal



DL dusumea

PIESE



DL

17

Dusumea larice
28x145 · larice 28x145



H4

~204

Inox 5x60 · 2 / joista



F1

2

Ulei tec · ulei

SCULE

 Bormasina

 Cui distantier 5 mm

 Pensula

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

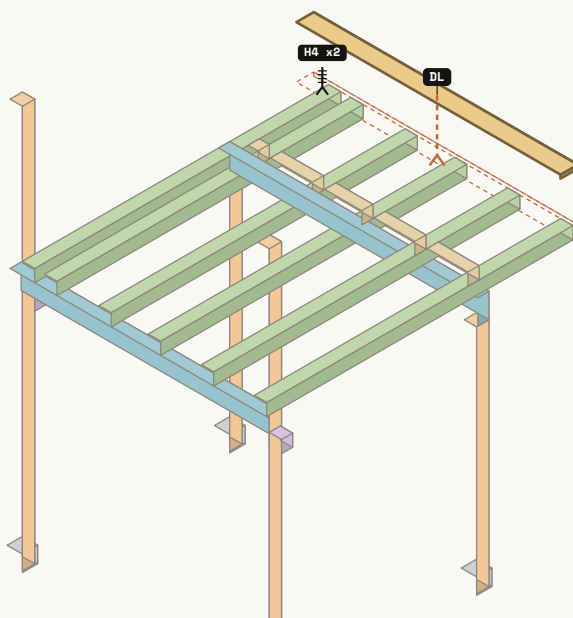
oblic

= la
unghi

Folosim **suruburi + buloane**, nu cuie.

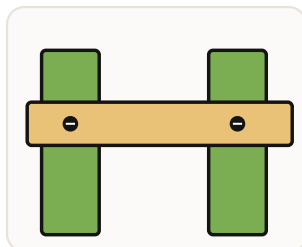
1

Aseaza prima scandura la marginea din fata, perpendicular pe joiste. 2 suruburi inox (H4) pe fiecare joista.

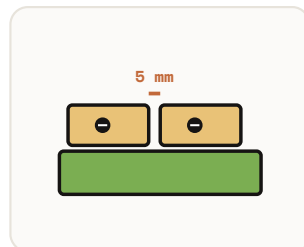


ZOOM · CUM SE PRINDE

DE SUS



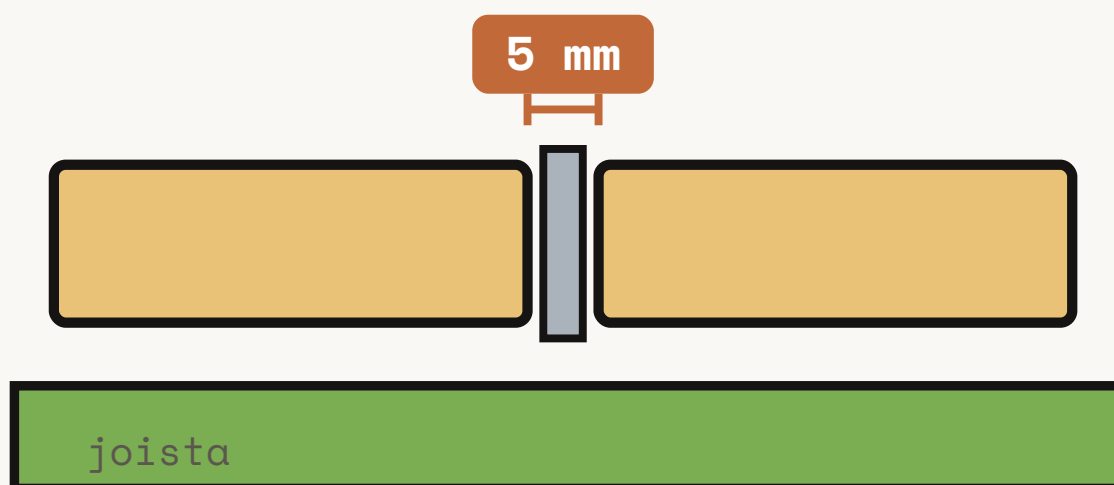
LATERAL



H4 x2 / joista · INOX · gol de 5 mm

2

Lasa 5 mm intre scanduri (pune un cui ca distantier). Apa se scurge, lemnul respira.

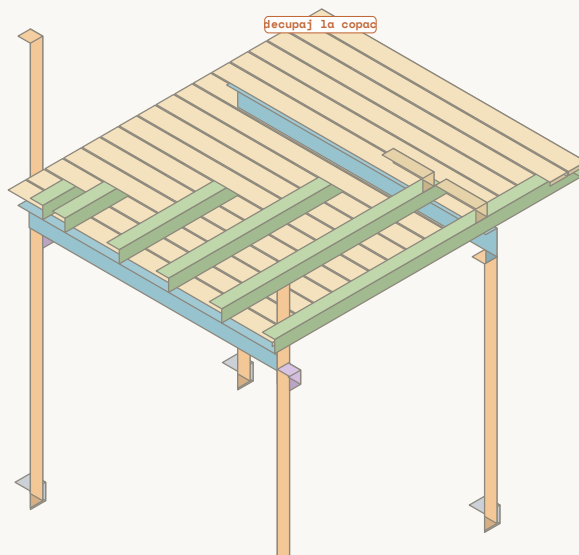


un cui ca distantier, apoi insurubezi

! CRITIC — Gol egal de 5 mm intre toate scandurile.

3

Continua pana acoperi tot — toate 17 scandurile ajung la grinda din spate. La copac, taie scandurile in jurul gaurii, cu joc.



4

Dupa montaj, da cu ulei (F1) pe toata dusumeaua de larice.



dusumea larice

GATA CAND

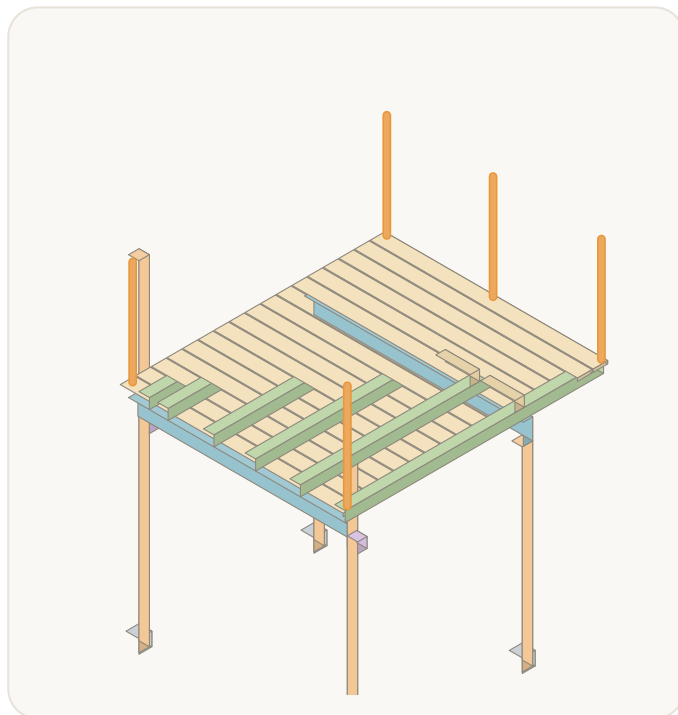
- ☐ Toate scandurile cu 2 H4 / joista
- ☐ Gol egal de 5 mm
- ☐ Decupaj la copac
- ☐ Uns cu ulei

FISA 11 / 11

Balustrada si poarta

~3-4 ore

CRITIC



ST stalp DL dusumea

PIESE



SB ~7
Stalpisor 58x58 ·
stalpisor 58x58, de cadru



SP ~24
Sipci balustrada ·
sipci, gol <9 cm



MC ~7 m
Mana curenta · mana
curenta sus



BM ~12
Bulon M10 · bulon M10 in
cadru

SCULE



Bormasina



Nivela



Fierastrau

PAS CU PAS



cap surub
(fata)



ascuns (pe
spate)



directie

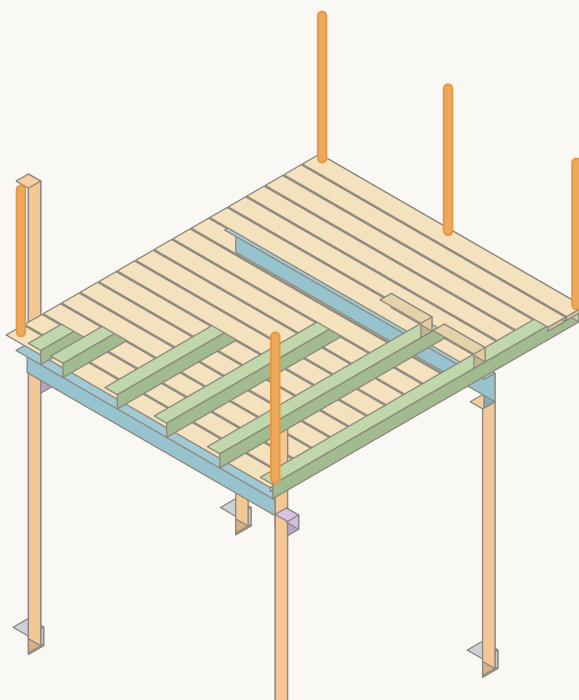
oblic

= la
unghi

Folosim **suruburi + buloane**, nu cuie.

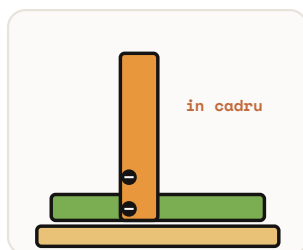
1

Monteaza stalpisorii balustradei (SB, 58x58), bulonati de CADRU (joiste/grinda), nu de dusumea. Jur-impregjur, inclusiv pe nasul consolei.

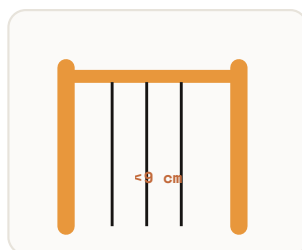


ZOOM · CUM SE PRINDE

LATERAL



FATA

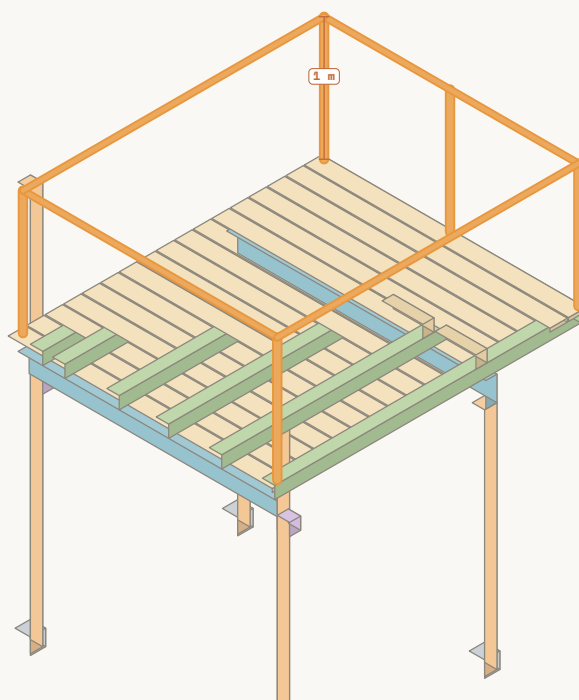


Stalpisor bulonat de CADRU · goluri sub 9 cm

! CRITIC — Stalpisorii se prind de cadru (joiste/grinda), nu de dusumea.

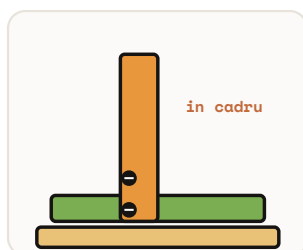
2

Pune mana curenta sus, la 1 m inaltime, pe tot conturul.

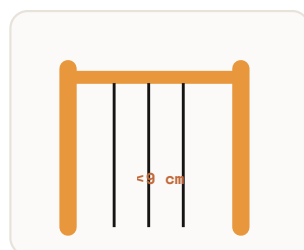


ZOOM · CUM SE PRINDE

LATERAL



FATA



Stalpisor bulonat de CADRU · goluri sub 9 cm

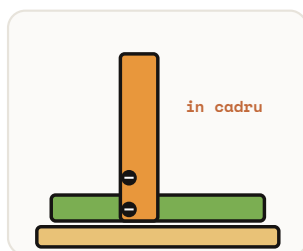
! CRITIC — Mana curenta exact la 1 m, pe tot conturul.

3

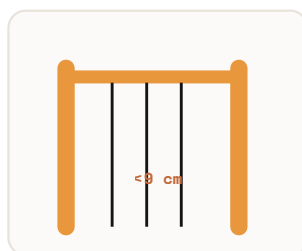
Umple cu sipci (SP), cu goluri sub 9 cm (un copil nu trebuie sa incapa sau sa se catere usor).

ZOOM · CUM SE PRINDE

LATERAL



FATA



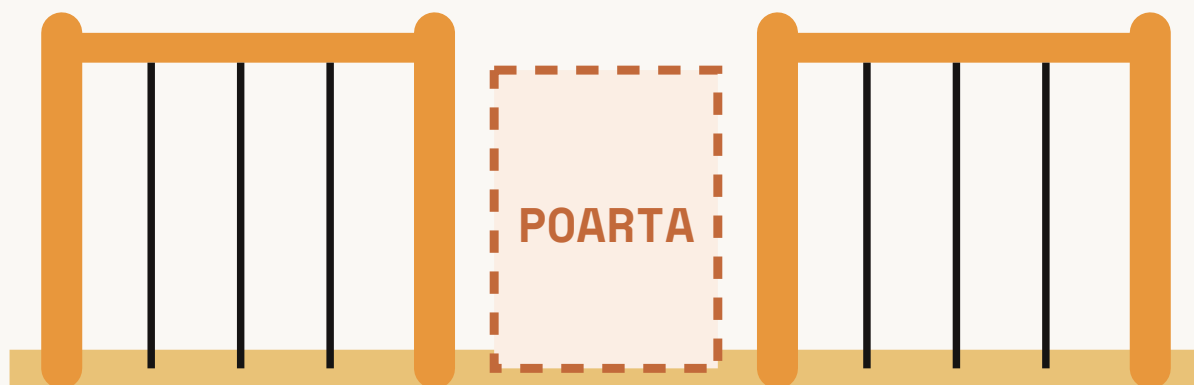
Stalpisor bulonat de CADRU · goluri sub 9 cm

! CRITIC — Goluri sub 9 cm peste tot.

4

Lasa o poarta pe partea S3 (opusa mesei-copac), unde urca scara. Testeaza balustrada impingand cu putere.

latura S3 (unde urca scara) – vedere din fata



travee fixa, goluri <9 cm

travee fixa

! CRITIC — Poarta pe S3; testeaza toata balustrada impingand cu putere.

GATA CAND

- ☒ Mana curenta la 1 m de la dusumea
- ☒ Goluri sub 9 cm peste tot
- ☒ Poarta pe S3
- ☒ Stalpisori bulonati de cadru, testati prin impingere

Siguranta & verificare

Cinci reguli care nu se negociaza — un copil va sta la 2,2 m.

Inaltime 2,2 m

Balustrada de 1 m jur-imperejur, goluri sub 9 cm, poarta pe partea S3. Sub platforma, sol moale (scoarta/nisip) pe ~5 m².

Copacul nu tine nimic

Corcodusul trece liber prin gaura cu joc de 3-5 cm. Blatul mesei se prinde de cadrul propriu, NICIODATA de copac.

Ordinea de siguranta

Proptelele raman pana montezi contravantuirile. Nimeni nu sta pe consola (balcon) inainte de contrafisa de sub nas.

Legaturi verificate

Coltare anti-ridicare la fiecare joista; coltar anti-smulgere la grinda din spate. Dupa montaj, reverifica strangerea buloanelor portante (polite + baza).

Lemn protejat

Grunduieste lemnul de structura INAINTE de montaj; ulei pe dusumea DUPA montaj. Capetele de stalp in papuc, cu drenaj, sa nu putrezeasca.